

Chapter 4a: Monotone Operators and Iterative Methods

Fixed Point Theory and Nonlinear Analysis

Based on Pathak's Introduction to Nonlinear Analysis

July 14, 2026

Chapter Overview

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- 2 Maximal Monotone Operators
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Definition: Monotone Operators

Definition (Monotone Operator)

Let $T : \mathcal{D}(T) \subset X \rightarrow X^*$ be an operator. Then:

(i) T is **monotone** if:

$$\langle Tx - Ty, x - y \rangle \geq 0 \quad \forall x, y \in \mathcal{D}(T)$$

(ii) T is **strictly monotone** if the inequality is strict for $x \neq y$

(iii) T is **strongly monotone** if $\exists c > 0$:

$$\langle Tx - Ty, x - y \rangle \geq c\|x - y\|^2 \quad \forall x, y \in \mathcal{D}(T)$$

(iv) T is **dissipative** if $-T$ is monotone

Remark: For complex Banach spaces, consider only the real part of $\langle Tx - Ty, x - y \rangle$.

Examples of Monotone Operators

Example

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a monotone increasing function. Then f is a monotone operator.

Example

Let $\mathcal{H}$ be a Hilbert space and $A : \mathcal{H} \rightarrow \mathcal{H}$ a compact self-adjoint linear operator. Then A is monotone if all eigenvalues are nonnegative.

Example

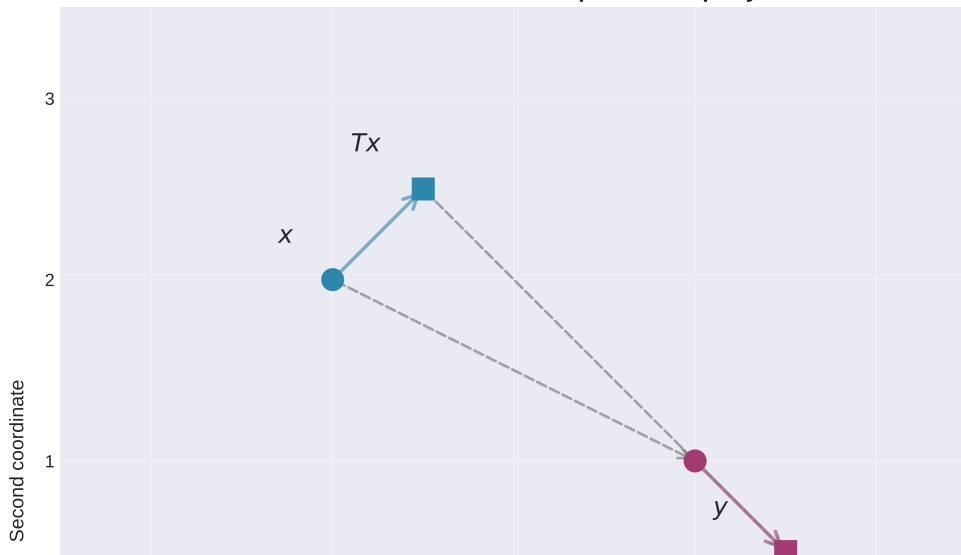
The differential operator $A : L^2[0, 1] \rightarrow L^2[0, 1]$ defined by $A = -\frac{d^2}{dt^2}$ with domain

$$\mathcal{D}(A) = \{x \in L_2[0, 1] : x'(t), x''(t) \in L_2[0, 1], x(0) = x(1) = 0\}$$

is a demi-continuous monotone operator.

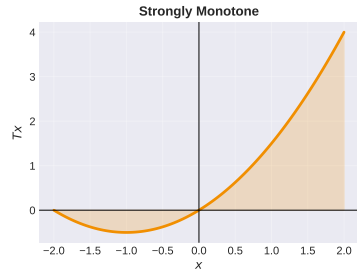
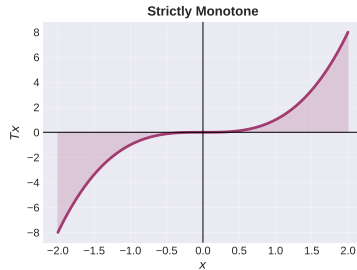
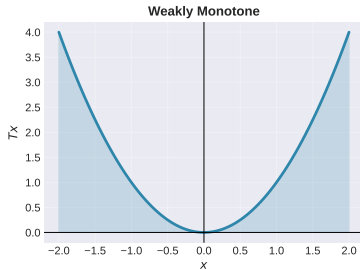
Monotone Operator Visualization

Visualization of Monotone Operator Property



Types of Monotonicity

Types of Monotonicity



- **Weakly monotone:** General monotonicity condition holds
- **Strictly monotone:** Strict inequality for $x \neq y$
- **Strongly monotone:** Controlled by distance with constant $c > 0$

Key Theorem on Monotone Operators

Theorem (Boundedness of Monotone Operators)

Let C be a convex subset of a Banach space X , and let $T : X \rightarrow X^$ be monotone. If the interior of $\mathcal{D}(T)$ is non-empty, then T is locally bounded at $x \in \text{int } \mathcal{D}(T)$.*

Theorem (Continuity and Monotonicity)

Let X be a Banach space and $T : \mathcal{D}(T) \subset X \rightarrow X^$ be monotone with $\mathcal{D}(T)$ dense in X . Then T is demi-continuous at $x \in \text{int } \mathcal{D}(T)$ if and only if it is hemi-continuous and locally bounded.*

Definition: Maximal Monotone Operator

Definition (Maximal Monotone Operator)

An operator $T : X \rightarrow 2^{X^*}$ is **maximal monotone** if:

- 1 T is monotone
- 2 There is no monotone operator $\tilde{T}$ with $T \subsetneq \tilde{T}$ (i.e., T is not a proper subset of any other monotone operator)

In other words, the graph of T cannot be extended while preserving monotonicity.

- Maximal monotone operators are fundamental in convex analysis and optimization
- They include subdifferentials of convex functions
- They have excellent regularity properties

Properties of Maximal Monotone Operators

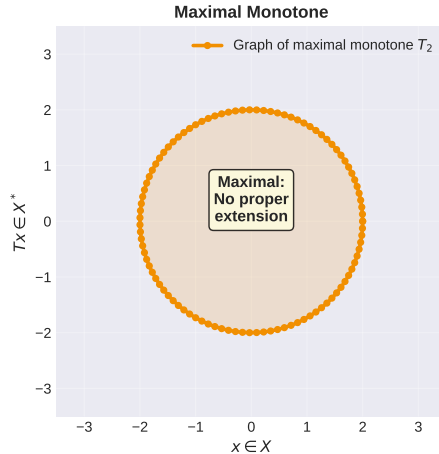
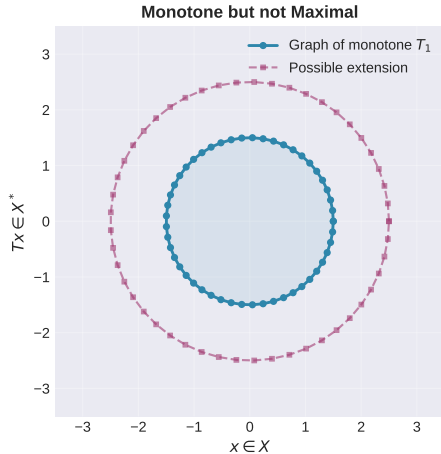
Theorem (Key Properties)

Let $T : X \rightarrow 2^{X^*}$ be a maximal monotone operator. Then:

- 1 The domain $\mathcal{D}(T) = \{x \in X : T(x) \neq \emptyset\}$ is convex
- 2 For every $\lambda > 0$, the operator $J_\lambda = (I + \lambda T)^{-1}$ is well-defined and nonexpansive
- 3 The graph $G(T)$ is closed in the weak topology
- 4 T is surjective in the sense that $\mathbb{R}(I + T) = X^*$

Visualization: Maximal Monotone Operators

Maximality of Monotone Operators



Maximality means the graph cannot be extended while preserving monotonicity.

Surjectivity for Maximal Monotone Operators

Theorem (Theorem 4.31: Surjectivity for Maximal Monotone)

Let $T : X \rightarrow 2^{X^}$ be a maximal monotone operator from a reflexive Banach space X into X^* . Further, let T be coercive with $0 \in \mathcal{D}(T)$. Then:*

$$\mathbb{R}(T) = X^*$$

Coercivity condition:

$$\frac{\|w\|}{\|x\|} \rightarrow \infty \quad \text{as } \|x\| \rightarrow \infty, \quad w \in Tx$$

This guarantees that T is surjective (onto).

Surjectivity Results (Continued) I

Theorem (Theorem 4.25)

Let X be a reflexive Banach space and B_0 an open ball containing the origin. If $T : X \rightarrow 2^{X^}$ is a maximal monotone operator satisfying certain boundedness conditions, then $\mathbb{R}(T) = X^*$.*

Theorem (Theorem 4.30)

Let X be a strictly convex reflexive Banach space. Let $T_1, T_2 : X \rightarrow 2^{X^}$ be maximal monotone operators. If $T_1 + T_2$ is surjective, then specific properties of the composition hold.*

Theorem (Theorem 4.32)

Let X be a reflexive Banach space and $T : X \rightarrow 2^{X^}$ maximal monotone. If certain growth conditions are satisfied, then T is surjective.*

Problem Formulation

Goal: Solve the operator equation

$$Tx = y, \quad y \in X$$

where $T : X \rightarrow X^*$ is an operator (possibly nonlinear and monotone).

Approximate Problem: For finite dimensional subspaces $X_n \subset X$:

$$T_n x_n = P_n y$$

where $P_n : X \rightarrow X_n$ is a projection operator.

Definition (Projective Solvability - PS-Solvable)

Equation (4.21) is PS-solvable if there exists $N > 0$ such that for each $n \geq N$ and given $y \in X$, equation (4.22) has a unique solution $x_n \in X_n$ with $x_n \rightarrow x$ in X , where x is the unique solution of (4.21).

Constructive Solvability Theorem

Theorem (Theorem 4.35: Approximate Solvability)

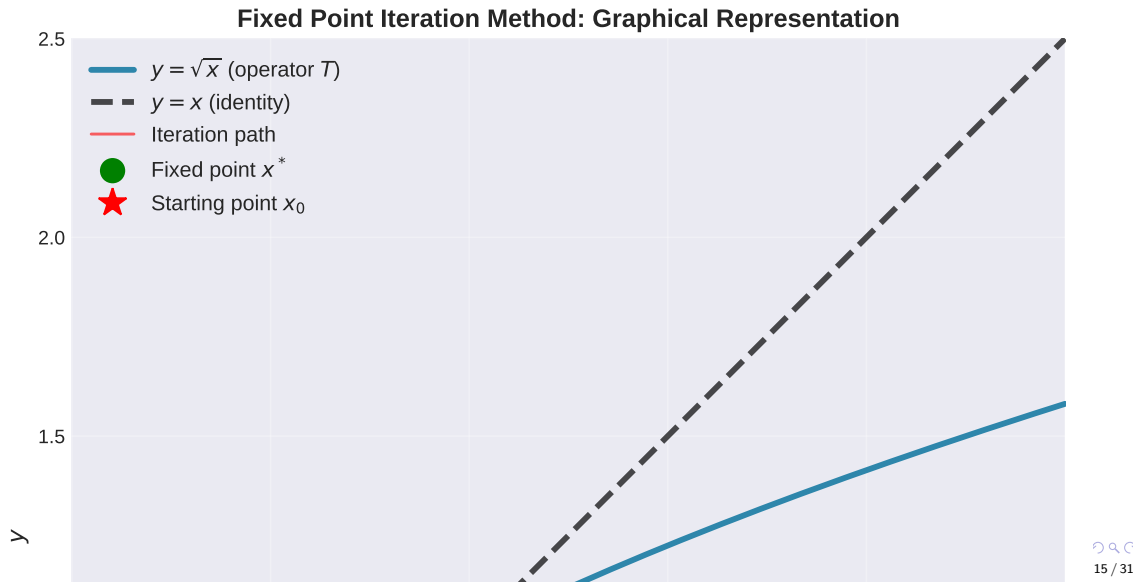
Suppose there exist a continuously monotonically increasing real function $\alpha(r)$ with $\alpha(0) = 0$ and $\lim_{r \rightarrow \infty} \alpha(r) = \infty$, and an integer $N > 0$ such that:

- (a) T_n is continuous in X_n for each $n \geq N$ and $T_n x \rightarrow GTx$ for each $x \in X$*
- (b) For each $n \geq N$ and all $x, y \in X_n$:*

$$\|T_n x - T_n y\| \geq \alpha(\|x - y\|)$$

Then for n large enough, (4.22) has a unique solution $x_n \in X_n$, and $x_n \rightarrow x$ as $n \rightarrow \infty$.

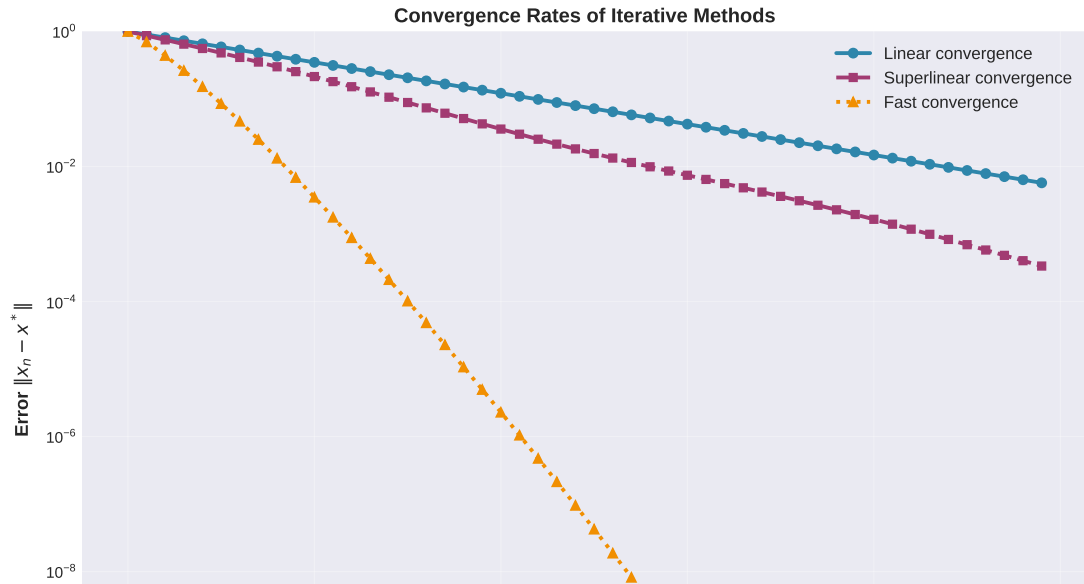
Fixed Point Iteration Scheme



Numerical Implementation

```
def fixed_point_iteration(T, y, x0, tol=1e-8, max_iter=1000):  
    """  
    Solve  $Tx = y$  using fixed point iteration  
     $x_{n+1} = x_n - \lambda(T(x_n) - y)$   
    """  
    x = x0  
    errors = []  
  
    for n in range(max_iter):  
        # Compute residual  
        residual = np.linalg.norm(T(x) - y)  
        errors.append(residual)  
  
        if residual < tol:  
            print(f"Converged in {n} iterations")  
            return x, errors  
  
        # Update:  $x_{new} = x - \alpha(T(x) - y)$   
        lambda_n = 1.0 / (n + 2) # Decreasing step size  
        x = x - lambda_n * (T(x) - y)  
  
    return x, errors
```


Convergence of Iterative Methods



Subdifferential of Convex Functions

Definition (Subdifferential)

Let $f : X \rightarrow \mathbb{R} \cup \{\infty\}$ be a convex function. The **subdifferential** of f at x is:

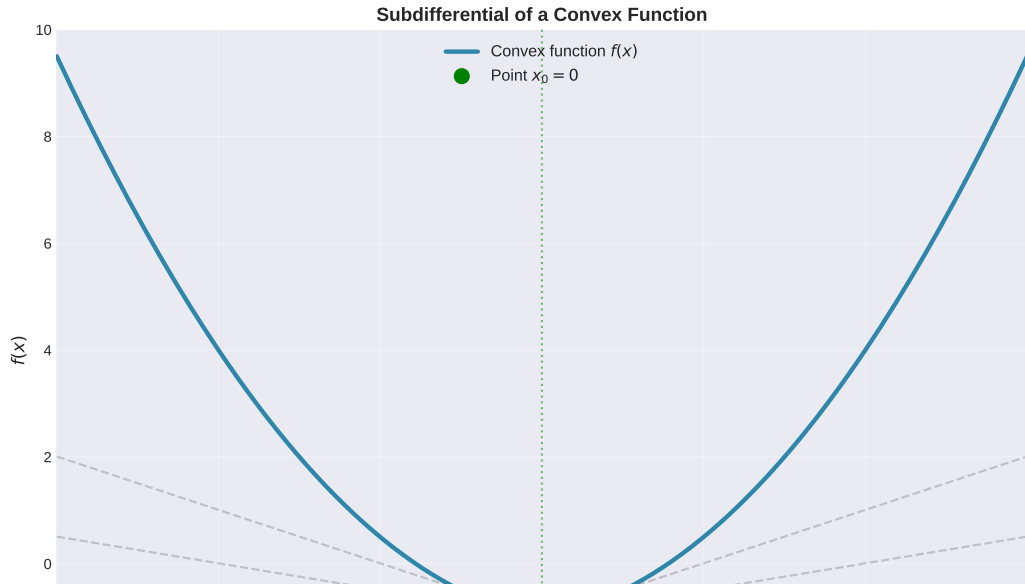
$$\partial f(x) = \{x^* \in X^* : f(y) \geq f(x) + \langle x^*, y - x \rangle \quad \forall y \in X\}$$

An element $x^* \in \partial f(x)$ is called a **subgradient**.

Key Property: The subdifferential operator $\partial f : X \rightarrow 2^{X^*}$ is:

- Monotone
- Maximal monotone in many cases
- Characterized by the support property

Subdifferential Visualization



Connection to Monotonicity I

Theorem

Let $f : X \rightarrow \mathbb{R} \cup \{\infty\}$ be a proper convex function. Then:

- 1 The subdifferential ∂f is a monotone operator
- 2 If f is strictly convex, then ∂f is strictly monotone
- 3 If f is strongly convex with modulus c , then ∂f is strongly monotone with the same modulus

Theorem (Maximal Monotonicity of Subdifferential)

If $f : X \rightarrow \mathbb{R} \cup \{\infty\}$ is a proper convex and lower semi-continuous function, then ∂f is a maximal monotone operator.

This fundamental result connects convex analysis with operator theory.

Definition: Pseudomonotone Operator

Definition (Pseudomonotone Operator)

An operator $T : X \rightarrow 2^{X^*}$ is **pseudomonotone** if:

- 1 For each $x \in X$, the set $T(x)$ is non-empty, closed, and convex
- 2 If $\{x_n\} \subset X$ with $x_n \rightharpoonup x$ and if $x_n^* \in T(x_n)$ with $x_n^* \rightharpoonup x^*$, then:

$$\limsup_{n \rightarrow \infty} \langle x_n^*, x_n - x \rangle \leq 0 \implies x^* \in T(x)$$

- 3 T is bounded: the image of bounded sets is bounded

Key differences from monotone:

- Monotone operators $\Rightarrow$ Pseudomonotone operators
- Pseudomonotone is weaker but still useful
- Applicable to broader class of problems

Properties and Examples I

Theorem (Properties of Pseudomonotone Operators)

If T is pseudomonotone, then:

- 1 *T is bounded (maps bounded sets to bounded sets)*
- 2 *For maximal pseudomonotone T , the equation $Tx = y$ has solutions under coercivity conditions*
- 3 *Many variational and semilinear problems lead to pseudomonotone operators*

Example

The Nemytskii operator from the space of functions:

$$N_f(x)(t) = f(t, x(t))$$

is pseudomonotone under appropriate growth and continuity conditions on f .

Definition: Accretive and φ -Accretive Operators

Definition (Accretive Operator)

An operator $T : X \rightarrow X^*$ is **accretive** if:

$$\langle Tx - Ty, \varphi(x - y) \rangle \geq 0 \quad \forall x, y \in \mathcal{D}(T)$$

where $\varphi : X \rightarrow X^*$ is the duality mapping.

Definition (Strongly φ -Accretive)

T is **strongly φ -accretive** if $\exists c > 0$:

$$\langle Tx - Ty, \varphi(x - y) \rangle \geq c \|x - y\|^2 \quad \forall x, y \in \mathcal{D}(T)$$

Relation to monotonicity:

- In Hilbert spaces: accretive $\leftrightarrow$ monotone
- In Banach spaces: accretive is a generalization
- Strongly φ -accretive $\Rightarrow$ unique fixed point

Surjectivity for φ -Accretive Operators

Theorem (Surjectivity for Strongly φ -Accretive)

Let $T : X \rightarrow X^$ be a continuous, strongly φ -accretive operator on a Banach space X . Then for every $y \in X^*$, the equation $Tx = y$ has a unique solution $x \in X$.*

Corollary (Range Characterization)

If T is maximal φ -accretive and coercive, then:

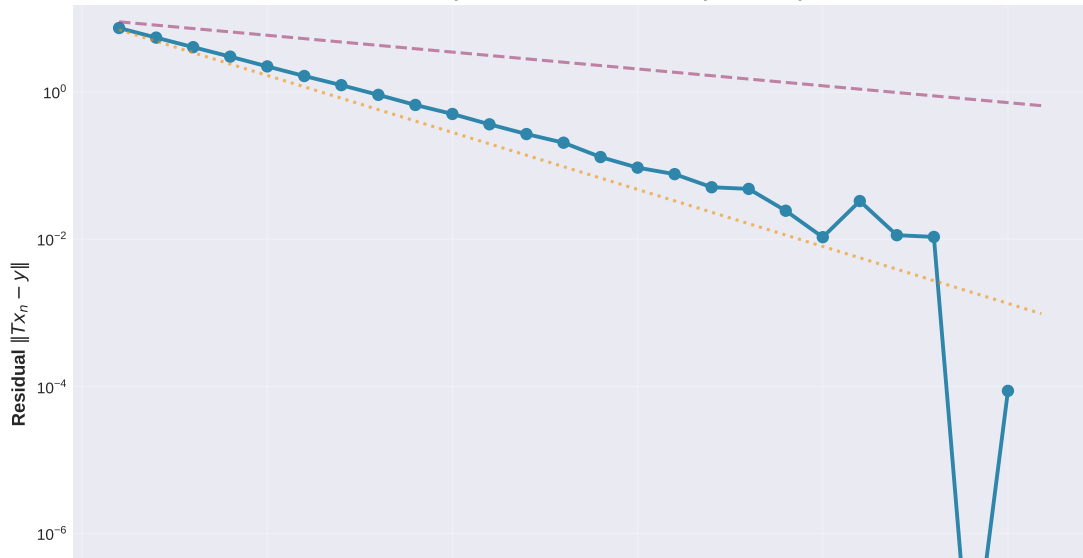
$$\mathbb{R}(T) = X^*$$

Proof Strategy:

- 1 Show T is injective (from strong monotonicity)
- 2 Show T is surjective (from coercivity)
- 3 Combine to get existence and uniqueness

Numerical Example: Convergence Analysis

Numerical Example: Iterative Solution of Operator Equation



Example: Solving a Nonlinear Equation

Problem: Solve $Tx = y$ where $T(x) = x^3 + 2x$ and $y = 3$

```
import numpy as np

def T(x):
    return x**3 + 2*x

def dT(x):
    """Derivative (approximation of monotonicity check)"""
    return 3*x**2 + 2

y = 3.0
x = 0.5 # Initial guess

for n in range(50):
    # Newton iteration
    residual = abs(T(x) - y)
    if residual < 1e-10:
        print(f"Solution found: x = {x:.10f}")
        break

    x_new = x - (T(x) - y) / dT(x)
    x = x_new
```

Solution: $x \approx 0.8436$ (verified by $T(0.8436) \approx 3.0$)

Example: Monotone Operator Properties

```
# Check monotonicity of  $T(x) = x^3 + 2x$ 
def check_monotonicity(x1, x2):
    Tx1 = x1**3 + 2*x1
    Tx2 = x2**3 + 2*x2

    # Compute inner product (in 1D: product)
    inner_prod = (Tx1 - Tx2) * (x1 - x2)

    return inner_prod >= 0 # Should be True for monotone

# Test points
test_pairs = [
    (0, 1), (0, 2), (1, 2), (-1, 1), (-2, -1)
]

print("Monotonicity check for  $T(x) = x^3 + 2x$ :")
for x1, x2 in test_pairs:
    is_mono = check_monotonicity(x1, x2)
    Tx1, Tx2 = x1**3 + 2*x1, x2**3 + 2*x2
    status = "YES" if is_mono else "NO"
    print(f"({x1:2}, {x2:2}): "
          f"(T({x1})-T({x2}))*(x1-x2) = "
          f"({Tx1-Tx2}*(x1-x2):7.1f) {status}")
```

Output shows the operator is monotone on all test pairs.

Key Takeaways

- **Monotone Operators:** Fundamental objects in functional analysis and optimization
 - Monotonicity: $(Tx - Ty, x - y) \geq 0$
 - Three types: weak, strict, strongly monotone
- **Maximal Monotone:** Cannot be extended while preserving monotonicity
 - Essential for guaranteeing surjectivity
 - Includes subdifferentials of convex functions
- **Constructive Methods:** Solve operator equations via projections
 - Approximate X -dimensional problem in X_n
 - Convergence under appropriate conditions
- **Extensions:** Pseudomonotone and φ -accretive operators
 - Weaker conditions for broader applicability
 - Important for PDE and variational problems

Monotone operators appear in:

- ➊ **Optimization:** Subdifferential of convex functions
 - Proximal methods: $x_{n+1} = \text{prox}_{\lambda f}(x_n)$
 - Forward-backward splitting
- ➋ **Partial Differential Equations:**
 - Variational formulation of elliptic problems
 - Existence proofs via monotone operator theory
- ➌ **Variational Inequalities:**
 - Find $x \in K$ s.t. $\langle Tx, y - x \rangle \geq 0$ for all $y \in K$
 - Applications: elasticity, contact problems
- ➍ **Fixed Point Theory:**
 - Relationship between monotonicity and fixed points
 - Iterative approximation of zeros

References and Further Reading

Key References:

- Pathak, H. K. (2018). An Introduction to Nonlinear Analysis and Fixed Point Theory. Springer.
- Brézis, H. (2011). Functional Analysis, Sobolev Spaces and Partial Differential Equations. Springer.
- Zeidler, E. (1990). Nonlinear Functional Analysis and Its Applications. Springer.
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- Rockafellar, R. T. (1970). Convex Analysis. Princeton University Press.

Topics for Further Study:

- Proximal point algorithm and monotone operators
- Operator splitting methods
- Monotone operators on Hilbert spaces
- Nonexpansive mappings and fixed points

Thank you!

Questions?